A3: Style Transfer Report

Student: Tyler Buckley

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Instructor: Dr. Shen

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Introduction

Neural Style Transfer (NST) combines the content of one image with the artistic style of another using deep neural networks. This project implements the method proposed by Gatys et al. (2016) using a pre-trained VGG19 model.

Part 1: Implementation

Content is extracted from conv4_2, and style is captured using Gram matrices across multiple layers.

Part 2: Experiments

Experiments varied style weights (5e4–1e7) and learning rates (0.001–0.003).

Results (Insert Images)

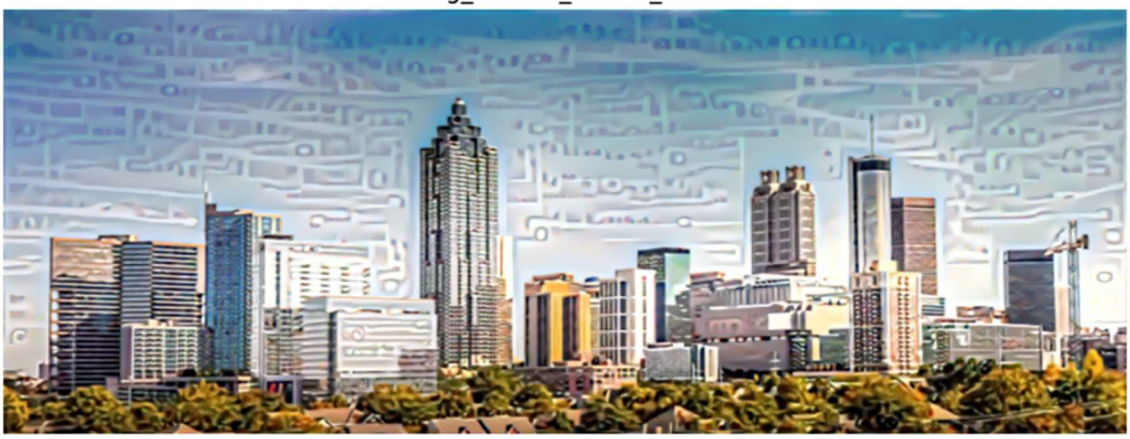
Insert:

- Content Image

- Style Image

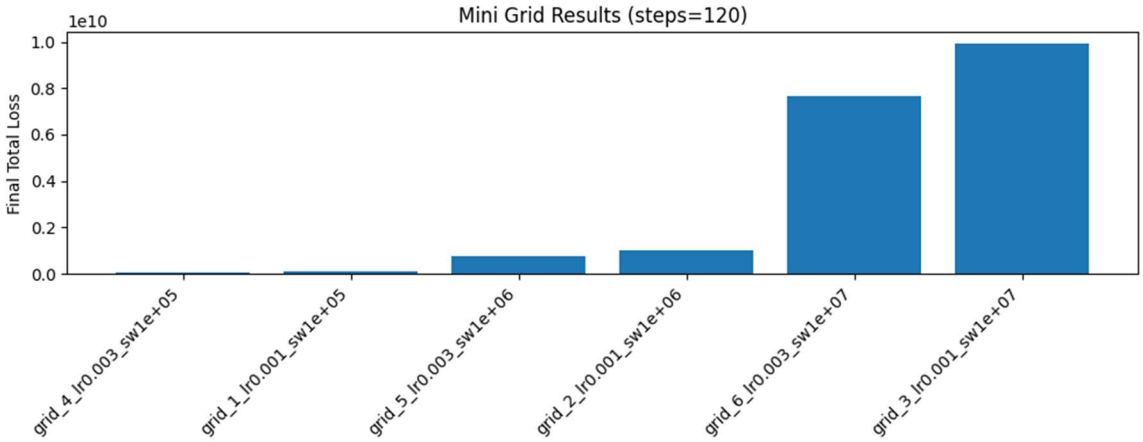
- Output Images

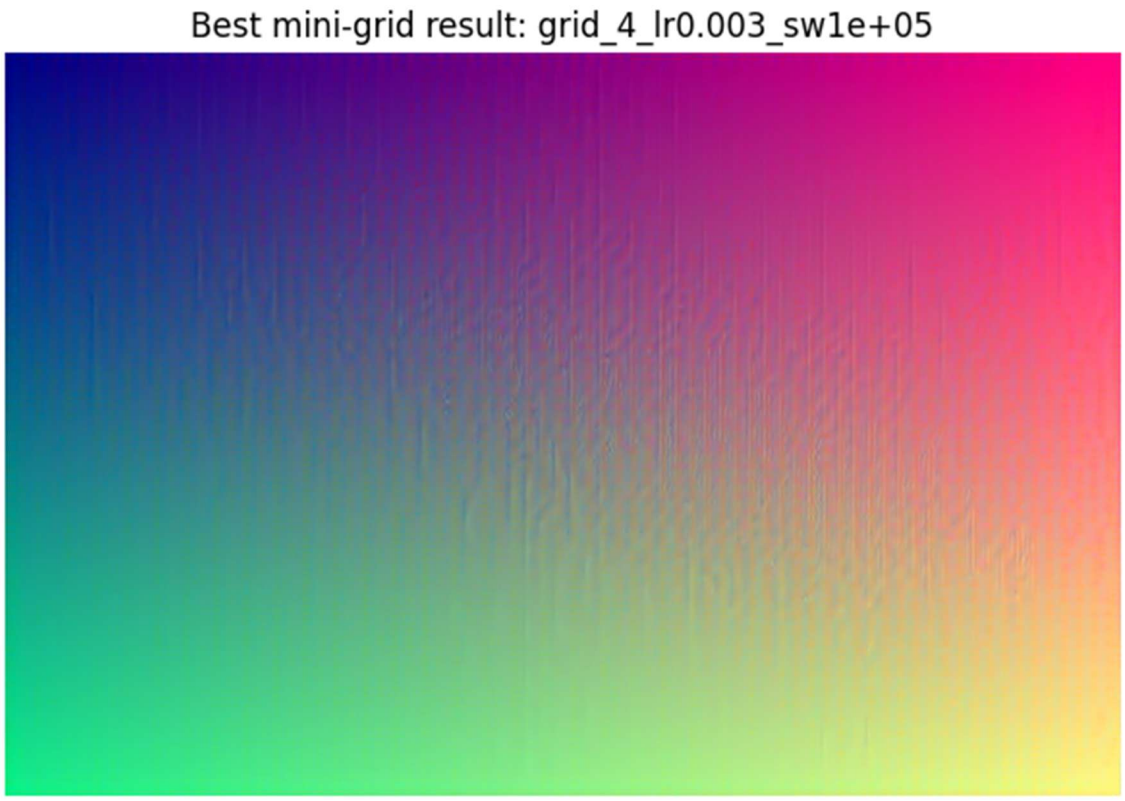
long_lr0.003_sw5e4_s600

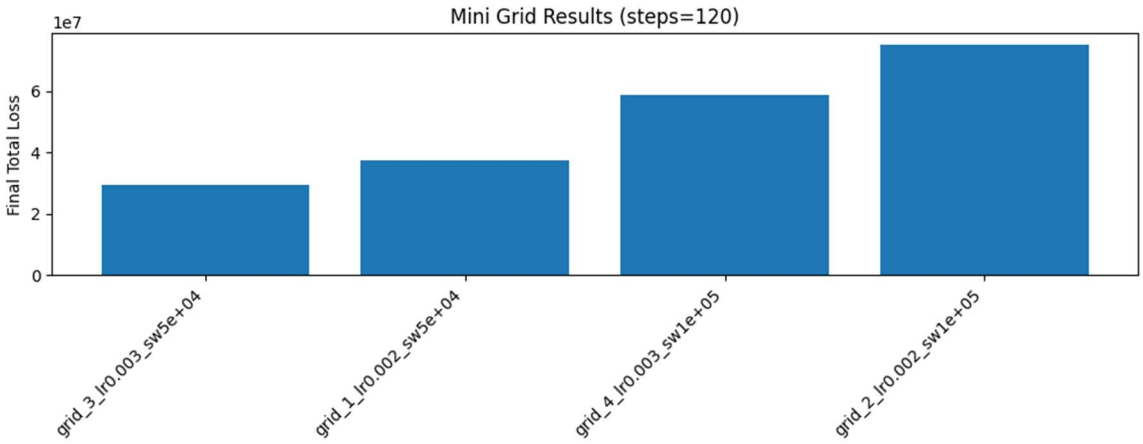


long_lr0.003_sw1e5_s600

- Graphs


Mini Grid Results (steps=120)
1e10
Final Total Loss
1.0
0.8
0.6
0.4
0.2
0.0
grid_4_lr0.003_sw1e+05
grid_1_lr0.001_sw1e+05
grid_5_lr0.003_sw1e+06
grid_2_lr0.001_sw1e+06
grid_6_lr0.003_sw1e+07
grid_3_lr0.001_sw1e+07


Best mini-grid result: grid_4_lr0.003_sw1e+05



Conclusion

Balanced hyperparameters produce the best visual results, preserving content while applying style.